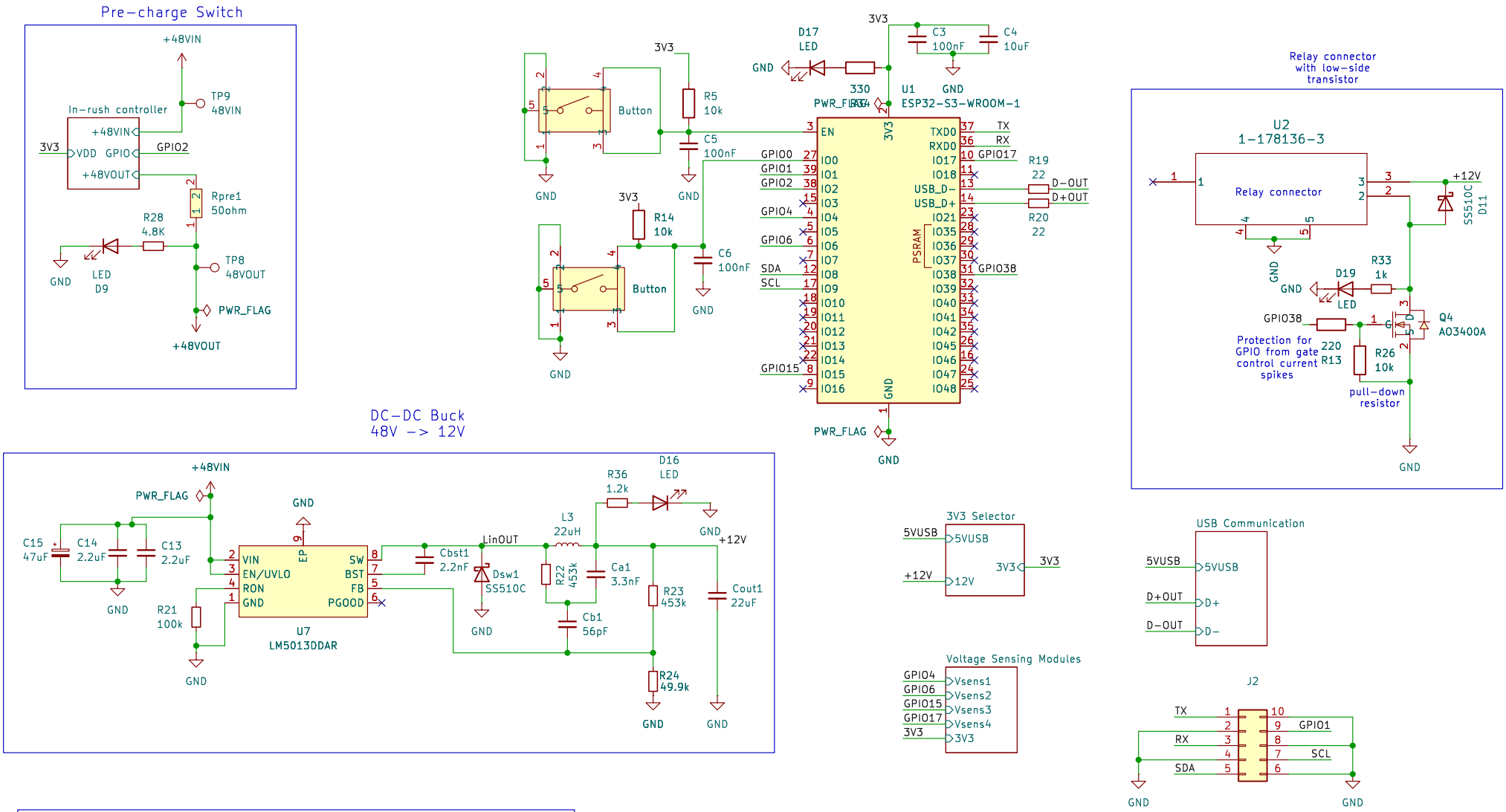
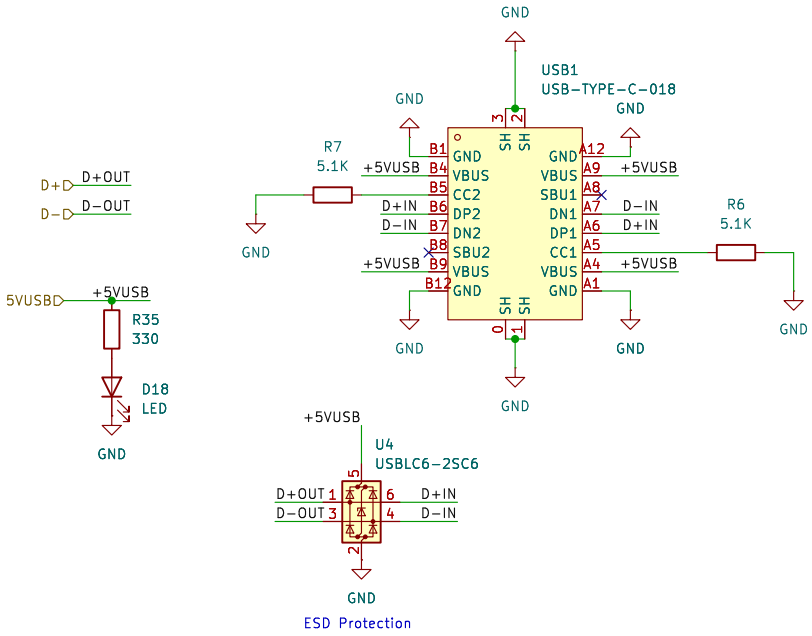
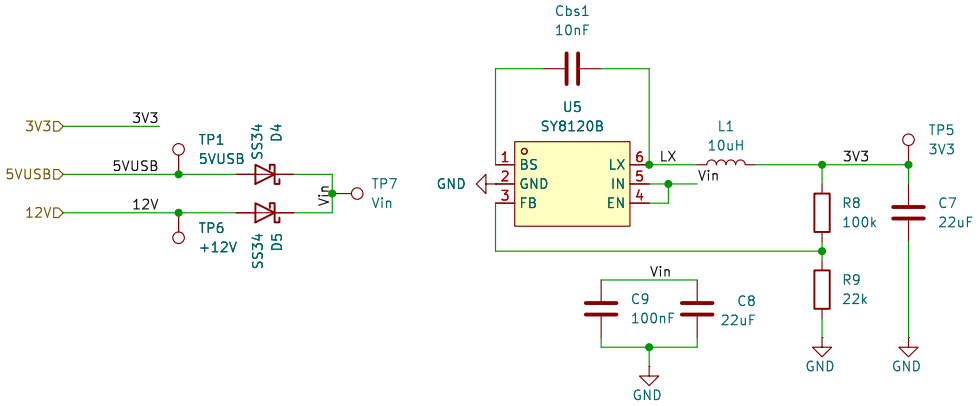

Pre-charge Switch
+48VIN
In-rush controller
TP9
48VIN
3V3
VDD
GPIO
GPIO2
+48VOUT
Rpre1
50ohm
R28
4.8K
GND
LED
D9
TP8
48VOUT
PWR_FLAG
+48VOUT
3V3
Button
R5
10k
C5
100nF
GND
D17
LED
C3
100nF
C4
10uF
330
U1
ESP32-S3-WROOM-1
PWR_FLAG
EN
3V3
GPIO0
GPIO1
GPIO2
GPIO4
GPIO6
SDA
SCL
GPIO15
TXD0
RXD0
GPIO17
IO17
IO18
USB_D-
USB_D+
IO21
PSRAM
IO35
IO36
IO37
GPIO38
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
TX
RX
R19
22
D-OUT
D+OUT
R20
22
R14
10k
C6
100nF
Button
Relay connector
with low-side
transistor
U2
1-178136-3
Relay connector
+12V
SS510C
D11
D19
LED
R33
1k
GPIO038
Q4
AO3400A
Protection for
GPIO from gate
control current
spikes
220
R13
R26
10k
pull-down
resistor
DC-DC Buck
48V -> 12V
+48VIN
PWR_FLAG
C15
47uF
C14
2.2uF
C13
2.2uF
R21
100k
U7
LM5013DDAR
VIN
EN/UVLO
RON
GND
EP
SW
BST
FB
PGOOD
Cbst1
2.2nF
LinOUT
Dsw1
SS510C
L3
22uH
R22
453k
Ca1
3.3nF
Cb1
56pF
R36
1.2k
D16
LED
R23
453k
R24
49.9k
Cout1
22uF
+12V
3V3 Selector
5VUSB
+12V
3V3
USB Communication
D+OUT
D-OUT
D+
D-
Voltage Sensing Modules
GPIO4
GPIO6
GPIO15
GPIO17
Vsens1
Vsens2
Vsens3
Vsens4
J2
GPIO1
SCL
XT30 IN/OUT
connectors
XT30PW-M
CN2
CN4
CN5
CN6
CN7
TP10
TP4
Mounting holes
H4
H3
H2
H1
Sheet: /
File: Good_Boy_Limiter.kicad_sch
Title: GBL (Good Boy Limiter)
Size: A4
Date: 2026-05-01
Rev:
KiCad E.D.A. 9.0.0
Id: 1/5



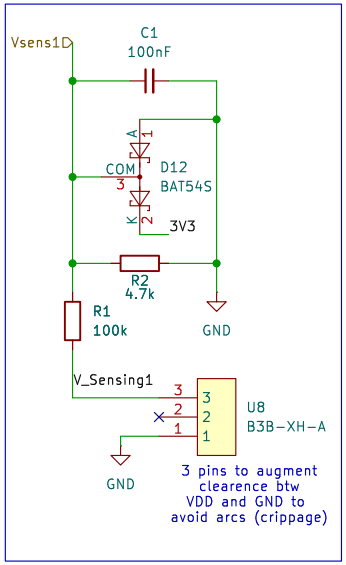
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Title: GBL		
Size: A4	Date: 2026-04-17	**Rev:**
KiCad E.D.A. 9.0.0		Id: 2/5



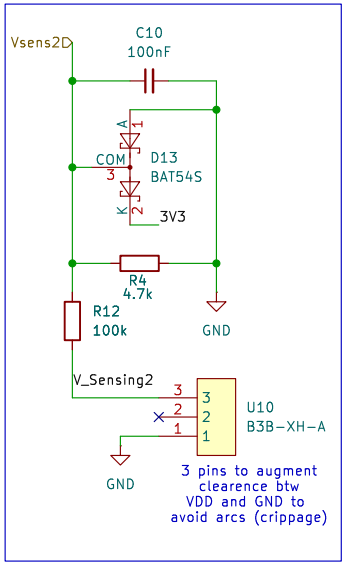
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.0		Id: 4/5

3V3 3V3

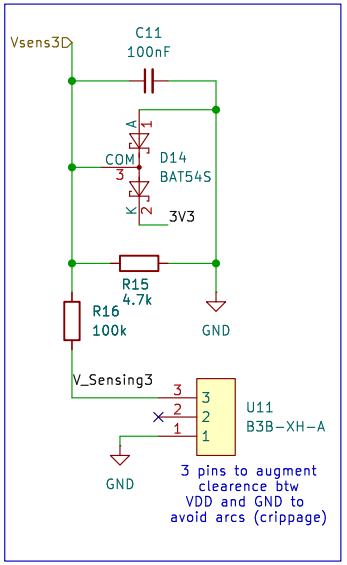
Resistor bridge
for sensing capacitor
charge



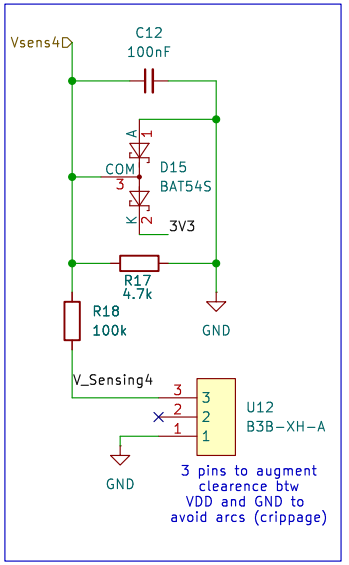
Resistor bridge
for sensing capacitor
charge



Resistor bridge
for sensing capacitor
charge



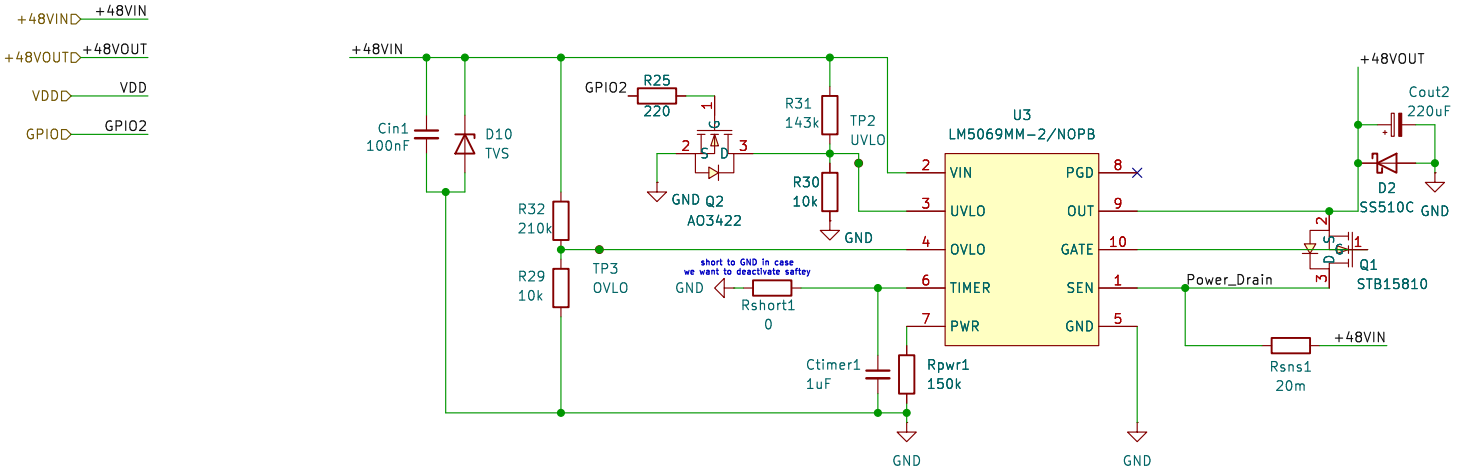
Resistor bridge
for sensing capacitor
charge



Sheet: /Voltage Sensing Modules/
File: vsensing_modules.kicad_sch

Title: GBL

Size: A4	Date: 2026-04-20	**Rev:**
KiCad E.D.A. 9.0.0		Id: 4/5



Sheet: /In-rush controller/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.0		Id: 5/5